RUTGERS–NEW BRUNSWICK
School of Arts and Sciences
Department of Physics and Astronomy



Eric Putney

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EDUCATION

Ph.D Candidate in Physics | *Rutgers, The State University of New Jersey* — August 2020 – Present

B.S *summa cum laude* Physics | *The University of New Mexico* — September 2016 – May 2020

RESEARCH FOCUS

Machine Learning for Galactic Dynamics — January 2021 – Present
PhD Thesis | Advisor: Dr. Matthew Buckley — *Rutgers | New Brunswick, New Jersey*

- Fully data-driven measurement of the local Milky Way's gravitational potential and dark matter density.
- Training normalizing flows to learn the six-dimensional kinematic phase space of nearby stars from Gaia DR3 to solve the general equilibrium collisionless Boltzmann equation.
- Developed novel technique for unbiasing the kinematic phase space of stars obscured by dust extinction.
- Developed non-parametric Jeans analysis for benchmarking our flows technique in a hydrodynamically-simulated galaxy.

Neutrino Flavor Dynamics in Core-Collapse Supernovae — October 2018 – May 2020
Undergraduate Honors Thesis | Advisor: Dr. Huaiyu Duan — *UNM | Albuquerque, New Mexico*

- Analyzed the evolution of dense neutrino gases evolving in supernovae.
- Compared numerical simulations of neutrino oscillations in dense matter to analytic predictions.

GBAR Positronium Excitation Laser — June 2019 – August 2019
NSF REU | Advisor: Dr. Pauline Comini — *CERN | Geneva, Switzerland*

- Calibrated the positronium excitation beam stabilization and thermal management, delivered beamline to the positronium chamber to produce neutral antihydrogren.
- Implemented remote instrumentation of the positronium excitation laser.

PRESENTATIONS AND PUBLICATIONS

Publications

- 2025 December | ClearPotential: Revealing Local Dark Matter in Three Dimensions | astro-ph 2512.09989 | NeurIPS ML4PS 2025
- 2024 December | Sweeping the Dust Away: Correcting the Phase Space Density of the Milky Way with Unsupervised Machine Learning | astro-ph 2412.14236
- 2023 May | Mapping Dark Matter in the Milky Way using Normalizing Flows and Gaia DR3 | JCAP, astro-ph 2305.13358
- 2022 May | Measuring Galactic Dark Matter through Unsupervised Machine Learning | MNRAS, astro-ph 2205.01129
- 2021 November | What Dark Matter Halos Tell Us About Dark Matter | Candidacy exam (*unpublished*)

- 2020 May | Numerical Analysis of Collective Neutrino Oscillations in Dense Neutrino Media. Honors thesis (unpublished)
- 2019 June | Measurement of the radial matrix elements for the $6s^2S_{1/2} \rightarrow 7p^2P_J$ transitions in cesium | Physical Review A, physics 1904.06362

Presentations

- 2026 May | Data Driven Methods for Measuring Galactic Dark Matter | Thesis Defense | Rutgers University, New Brunswick
- 2026 May | Efficient PINN Models of Cosmic Rays | Pheno 2026 Symposium, University of Pittsburgh. Abstract, Slides
- 2025 December | ClearPotential: Learning the Dust-Corrected Potential of the Milky Way Using Flows | Poster | NeurIPS ML4PS 2025 | Abstract, Letter, Poster
- 2025 July | Gaia and Dark Matter | Cargese Summer School | IESC | Indico
- 2025 April | Model-free analyses of Gaia: mapping the local potential, dust, and dark matter | YITP Seminar | Stony Brooks University | Abstract
- 2025 March | Model-free analyses of Gaia: mapping the local potential, dust, and dark matter | Galaxy Hour | The Ohio State University | Abstract
- 2025 March | Machine Learning and Galactic Dynamics | Graduate Seminar | University of Cincinnati
- 2024 November | Sweeping the Dust Away: An Unbiased Map of the Milky Way's Dark Matter and Gravitational Potential with Unsupervised Machine Learning | Abstract, Slides, Recording
- 2024 May | Sweeping the Dust Away: An unbiased map of the Milky Way's gravitational potential using unsupervised ML | DPF-PHENO 2024, University of Pittsburgh | Abstract, Slides
- November 2023 | Mapping Dark Matter in the Milky Way using Normalizing Flows and Gaia DR3 | ML4Jets 2023, DESY. Abstract, Slides
- 2023 May | Measuring Galactic dark matter through unsupervised machine learning | May 2023, Phenomenology 2023 Symposium, University of Pittsburgh. Abstract, Slides
- 2023 January | Measuring Galactic Dark Matter through Unsupervised Machine Learning | 241st AAS Meeting, Seattle WA. Abstract
- 2021 November | What Dark Matter Halos Tell Us About Dark Matter | PhD Candidacy Exam, Rutgers, the State University of New Jersey | Slides
- 2020 April | Numerical Analysis of Collective Neutrino Oscillations in Dense Neutrino Media | Undergraduate Honors Thesis, University of New Mexico | Slides, Poster
- 2019 August | Laser Control for the Production of Excited Positronium in GBAR | CERN Summer School | Slides
- 2019 April | Numerical Analysis of Collective Neutrino Oscillations in Core-Collapse Supernovae with Multidimensional Models | University of New Mexico Physics Day | Slides
- 2018 October | Precision Measurement of the $6S \rightarrow 7P_{1/2}$ Cesium Transition Radial Matrix Element via Simultaneous Absorption Spectroscopy | 2018 APS Four Corners, University of Utah | Abstract, Slides
- 2017 October | Analysis of Diffusion of a Rhodium Adatom on a Tungsten (111) Surface. | 2017 APS Four Corners, Colorado State University | Abstract, Poster

SCHOLARSHIPS AND AWARDS

- 2021 Rutgers Academic Fellowship
- 2020 Eion Gray Scholarship
- 2020 UNM Feynman Award
- 2019 Goldwater Scholar
- 2019 Rayburn Outstanding Student in Laboratory Physics and Astronomy

SERVICE

Rutgers Chapter of the AAUP-AFT
Graduate Worker Organizer

- Physics and Astronomy Department Representative — December 2022 – Present
- Fellows Organizing Committee Chair — January 2024 – December 2025
- Graduate Steering Committee and Executive Council Member — September 2024 – Present
- Short Term Staff Organizer for Clinically Focused University Practitioners — January 2025 – June 2025

Rutgers Physics & Astronomy Graduate Student Organization
President — September 2023 – June 2024

Rutgers Graduate Student Association
Treasurer — January 2025 – June 2025

REFERENCES

Dr. Matthew Buckley
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Dr. David Shih
Rutgers, The State University of New Jersey, New High Energy Theory Center
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